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| **Fiscal Year:** | FY 2004 | **Task Last Updated:** | FY 09/01/2004 |
| **PI Name:** | Phillips, William D. Ph.D. | | |
| **Project Title:** | Condensate Laboratory Aboard the Space Station (CLASS) | | |
| **Division Name:** | Physical Sciences | | |
| **Program/Discipline:** | FUNDAMENTAL PHYSICS | | |
| **Program/Discipline--Element/Subdiscipline:** | FUNDAMENTAL PHYSICS--Laser cooled atomic physics | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **Project Type:** | Ground | **Solicitation / Funding Source:** | 00-HEDS-02 |
| **Start Date:** | 10/01/2000 | **End Date:** | 10/04/2004 |
| **No. of Post Docs:** | 6 | **No. of PhD Degrees:** | |
| **No. of PhD Candidates:** | 4 | **No. of Master' Degrees:** | |
| **No. of Master's Candidates:** | 1 | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | 2 | **Monitoring Center:** | NASA JPL |
| **Contact Monitor:** | | **Contact Phone:** | |
| **Contact Email:** | | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Helmerson, Kristian ( NIST )<br>Lett, Paul ( NIST )<br>Porto, James ( NIST ) | | |
| **Grant/Contract No.:** | None | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | not received | | |

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| **Task Description:** | The ultralow temperatures and velocities achievable with Bose-Einstein condensation can be most fully exploited in a microgravity environment. Long interaction times, the manipulation of atoms with very small forces, and macroscopic quantum wave functions will be possible. To develop a Bose-Einstein condensate for use aboard the International Space Station. Experiments to be performed in the laboratory will develop the basic tools and experimental techniques needed to perform experiments in space. These may include matter wave interferometry with coherent matter wave, quantum reflection, and creation of picokelvin atom samples for future generation clocks. The combination of microgravity with Bose-Einstein condensation will allow the exploration of matter waves at extremely low temperatures and long (macroscopic) wavelengths. The research will enhance our understanding of BEC phenomena, and pave the road for the development of future technologies based on matter waves. In particular, it should be possible to create measurement instruments whose quality far surpasses that available today with earthbound techniques. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | Matter wave interferometers are intrinsically more sensitive to certain influences, like rotation, than are optical interferometers, all other instrument parameters being equal. Rotation and acceleration sensing is essential to measurements for such applications as inertial navigation, gravitational mapping (which can be used in applications like navigation, prospecting, and searching for underground cavities. Ground based work on matter wave interferometers will impact both space and earth implementations. Spin off applications of condensates include quantum information, a subject with important national security implications, both for decryption and encryption. Applications of Bose-Einstein condensates to atomic clocks have potential earth based applications in navigation and communication. |
| **Task Progress:** | Four separate experiments in the Laser Cooling and Trapping Group at NIST provide support to the task. Two of these specifically perform experiments with Bose-Einstein condensate, a third is being converted to operation with a BEC, and a fourth is developing tools for manipulation of condensates. During the past year, all of these experiments have been moved to the new Advanced Measurement Laboratory, recently completed on the NIST-Gaithersburg campus. We look forward to improved conditions and capabilities in our new lab space.<br>The first apparatus produces a Bose-Einstein condensate (BEC) of sodium atoms, and is focused primarily on the application of condensates as a coherent source of matter-waves, a main task objective. In previous years we have developed optical techniques for the coherent manipulation of condensates and applied them for interferometric and other phenomena requiring coherent matter waves. Many of these techniques have been adopted as standard practice in other laboratories. Among these are diffraction of a BEC, in which two or more intersecting laser beams forming a standing wave are applied to the BEC as a short pulse, splitting the BEC into multiple, coherent momentum states, separated by multiples of the differences in photon momenta. Another is Bragg diffraction, in which laser beams of different frequencies induce a coherent transition between the BEC at rest and a single new momentum. A set of nterfering laser beams also constitute an optical lattice (either moving or stationary) and we have investigated the behavior of atoms in such lattices, using the perspective of band structure and Bloch states borrowed from solid state physics. In this way we have created a “Block accelerator” for atoms, allowing the coherent transfer of large amounts of momentum to the atoms.<br><br>Using Bragg diffraction we have created time-domain interferometers that separate and recombine a condensate, and we have observed constructive and destructive interference of the entire condensate, or of portions of it, by manipulating the phase of the condensate in one path of the interferometer relative to the other. We have used this interferometer to observe non-linear effects in the condensate, that is, phase shifts caused by the atom-atom interactions.<br><br>We have recently explored two different kinds of multiple path interferometers. In one of these, we split the condensate into two momentum components with Bragg diffraction, allow the components to spatially separate, then spilt each again and again with a similar Bragg pulse. With equal times between successive splittings, some paths overlap and interfere. This process realizes a quantum random walk with atoms. Although the quantum random walk (QRW) is predicted to give a speed up over the classical random walk for certain contrived algorithms, the usefulness of the QRW in computing remains to be determined. If our experiment is successful, it may provide additional motivation and insight to find a practical application of the QRW. Since the falling of our condensate atoms in the vacuum chamber limits the number of walk steps (interferometer paths) that we can perform, the usefulness of the QRW, with our atomic system, would ultimately rely on an environment with significantly reduced gravity, such as that aboard the space station. The present status of the experiments is that we have observed non-classical distributions in the random walk, but cannot take many steps because of phase instabilities due to vibrations. We hope to improve on the vibration control. Of course, a space experiment could provide a much quieter vibration environment.<br><br>In the second multiple path interferometer we used successive diffraction pulses to split the condensate into multiple momentum components. The time between pulses does NOT allow the condensate components to separate spatially (near-field regime) and the interference after multiple steps is more easily seen. The process is similar to a quantum kicked rotor, and this is the first such experiment with a condensate. Our preliminary results have shown the effects of predicted (but until now, unobserved) higher order resonances in the momentum transfer, and we will continue with these experiements once the apparatus is operating in the new laboratory.<br><br>The second apparatus produces a BEC of rubidium atoms, which can be subsequently loaded into a various kinds of optical lattices. The apparatus is directed towards the study of quantum phase transitions to produce “non-classical” states of matter-waves, and studying condensates in lower dimensions. Loading the atoms into a 3D lattice, we have been able to see a quantum phase transition (a Mott-insulator transition) which create number states of atoms in an optical lattice; that is, there is a fixed number of atoms in each occupied site in the lattice. One important feature of such states is the possibility that they may be used to do sub-shot-noise atom interferometry, just as non-classical (squeezed) states of light allow sub-shot-noise optical interferometry. In this way, these experiments support the matter wave interferometry efforts. The experiments are also relevant to quantum information with neutral atoms as qubits.<br><br>We have also produced an array of 1-D condensates by applying a 2D lattice to the BEC, and have shown that the atoms in the 1-D trapping potential are highly correlated, exhibiting fermionic behavior due to the atoms’ repulsive interactions. The development of BEC in lower dimensions may facilitate the production of atom number states because of the strong correlation. We look forward to experiments in reduced gravity environment which should help in creating condensates with large numbers of atoms in lower dimensions and other topologies, and uniform distributions of atoms in number states. |

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| | In recent experiments in low dimension we have observed anomalous damping of the oscillatory motion, an unexpected feature that has surprised theorists and is currently under investigation both experimentally and theoretically in our laboratory.<br><br>We are in the process of setting up a third BEC experiment (with sodium) to investigate collisions of condensate atoms and coherent molecule formation. Collisions of BEC atoms can be both advantageous and disadvantageous for matter-wave optics. Coherent collisions can produce interesting non-linear matter-wave phenomena, such as 4-wave mixing and solitons (which we have observed in our laboratory) analogous to non-linear optical phenomena. Collisions, however, can also produce unwanted phase shifts as well as incoherent coupling to different momentum states, all of which may compromise atom interferometry. The new experiment builds on our previous experience with collisions of ultracold atoms. It has demonstrated trapping and evaporative cooling of atoms in a magnetic trap as well as collisions and molecule formation in the magnetic trap. With the arrival of a new vacuum chamber we expect to achieve BEC.<br><br>A fourth experiment does not itself involve BEC, but is aimed at creating correlated photon pairs that could be used to produce correlations in the BEC. The experiments aims to produce the correlated photons by degenerate four wave mixing in Rb vapor. Four wave mixing has been seen, electromagnetically induced transparency has been introduced to improve the extraction of the generated beams and has been shown to coexist with the four wave mixing process. We have yet to see the photon correlations (an observation which, if successful, will itself be a “first”) buthope to do that once some light scattering problems are addressed. |
| **Bibliography Type:** | Description: (Last Updated: 06/01/2005) |
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| **Awards** | Lett, Paul D. "Election as a Fellow of the Optical Society of America" Oct-2004 |
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| **Awards** | Porto, J. "Sigma Xi Young Scientist Award for Excellence in Scientific Research" May-2004 |
| **Awards** | Phillips, W. "Election as an Honorary Member of the Optical Society of America" Oct-2004 |
| **Presentation** | J. V. Porto "Transport of 1D Bosons in an optical lattice" International Conference on Atomic Physicis, Rio de Janeiro<br>Jul-2004 |
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| **Presentation** | K. Helmerson "Coherent Matter Wave Optics" symposium for JICQuP, University of Maryland, College Park, MD<br>Jan-2004 |
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| **Presentation** | K. Helmerson "Non-linear Atom Optics" IX Jorge André Swieca Summer School of Quantum and Nonlinear Optics, Belo Horizonte, Minas Gerais<br>Feb-2004 |
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| **Presentation** | W. Phillips "Almost Absolute Zero: the story of laser cooling and trapping" public lecture, Bates College, Maine<br>Oct-2003 |
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| **Presentation** | W. Phillips "A Bose Condensate in an Optical Lattice: cold atomic gases meet solid state physics" Physics Dept. Colloquium,, McMaster University, Canada<br>Oct-2003 |
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